

HOMEWORK 2

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Problem 1. Let $b > 1$.

- (a) If m, n, p, q are integers, $n > 0, q > 0$, and $r = \frac{m}{n} = \frac{p}{q}$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

- (b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.
(c) If $x \in \mathbb{R}$, define

$$B(x) = \{b^t : t \in \mathbb{Q}, t \leq x\}.$$

Prove that $b^r = \sup B(r)$ when $r \in \mathbb{Q}$, and use this to define

$$b^x = \sup B(x) \text{ for every } x \in \mathbb{R}.$$

- (d) Prove that $b^{x+y} = b^x b^y$ for all real x, y .

Proof. A. We are given that $\frac{m}{n} = \frac{p}{q} = r$, and an arbitrary real number $b > 1$. We can clearly deduce that $b = b$. Now raise b to the power of r for both sides: $b^r = b^r$. Now, for the LHS, let $r = \frac{m}{n}$, and $r = \frac{p}{q}$ for the RHS: $b^{m/n} = b^{p/q}$. Now, through exponent rules, we arrive at the conclusion that $b^r = (b^m)^{1/n} = (b^p)^{1/q}$. $\square$

Proof. B. We are given that r and s are rational numbers, and are asked to prove the statement $b^{r+s} = b^r b^s$. Let $r = \frac{m}{n}$ and $s = \frac{p}{q}$, where $m, n, p, q \in \mathbb{Z}, n, q \neq 0$. We can use this to show that $r + s = \frac{mq+np}{nq}$. Now, raise b^{r+s} , which may be denoted as $b^{\frac{mq+np}{nq}}$. We can deduce that $mq + np$ is addition between two integers. So, let $\alpha = mq$ and $\beta = np$, where $\alpha, \beta \in \mathbb{Z}$. We must show that $b^{\alpha+\beta} = b^\alpha b^\beta$. Assuming $\alpha, \beta \geq 0$, we can illustrate their equivalence:

$$\underbrace{b \cdot \dots \cdot b}_{\alpha+\beta \text{ } b's} = \underbrace{\overbrace{(b \cdot \dots \cdot b)}^{\alpha \text{ } b's} \overbrace{(b \cdot \dots \cdot b)}^{\beta \text{ } b's}}_{\alpha+\beta \text{ } b's}$$

therefore, $b^{\alpha+\beta} = b^\alpha b^\beta$. If either α or $\beta < 0$, then we can represent this as $b^{-\alpha} = \frac{1}{b^\alpha}$. We can use this result now to show that $(b^\alpha b^\beta)^{\frac{1}{nq}}$. We can do this because of our result in **A**. Now, we can distribute the power into the b 's, giving us $b^r b^s$. Thus, $b^{r+s} = b^r b^s$. $\square$

Proof. C. We are given the set $B(x) = \{b^t : t \in \mathbb{Q}, t \leq x\}$, and we are trying to prove when our input $r \in \mathbb{Q}$, then $b^r = \sup B(x)$. We may notice that $t \leq x \implies b^t \leq b^x$. The first part of this question is quite trivial, since we let $x = r$, which is some rational, and since t is also rational, $t = x = r$, which gives us b^r as our supremum, since, as deduced before, $b^t \leq b^x = b^r \forall t \in \mathbb{Q}$. Now we are given an $x \in \mathbb{R}$. We have already proven that rationals are supremum for $B(x)$, so now we must focus on if x is irrational. $\square$

Proof. D. Through **B**, we've shown that if x, y are both rationals, then this works out. Therefore, let's assume that x or y are irrational. We can deduce that b^{x+y} is a continuous function. Let $x_n = \sum_{i=0}^n \frac{\alpha_i}{10^i}$ and $y_n = \sum_{i=0}^n \frac{\beta_i}{10^i}$. We see that this is a partial sum, meaning that these are rational numbers. Therefore, b $\square$

Problem 2. If $z \in \mathbb{C}$ satisfies $|z| = 1$ (that is, $z\bar{z} = 1$), compute

$$|1 + z|^2 + |1 - z|^2.$$

Proof. We know $z = x + iy$ and $\bar{z} = x - iy$ for $x, y \in \mathbb{R}$, and that $|z| = \sqrt{x^2 + y^2}$. Now, we have $|1 + x + iy|^2 + |1 - x - iy|^2 = (1 + x)^2 + y^2 + (1 - x)^2 + y^2 = 2 + 2x^2 + 2y^2 = 2(1 + x^2 + y^2) = 2(1 + |z|^2)$. $\square$

Problem 3. Suppose $k \geq 3$, $x, y \in \mathbb{R}^k$, $|x - y| = d > 0$, and $r > 0$. Prove:

- (a) If $2r > d$, there are infinitely many $z \in \mathbb{R}^k$ such that $|z - x| = |z - y| = r$.
- (b) If $2r = d$, there is exactly one such z .
- (c) If $2r < d$, there is no such z .

How must these statement be modified if $k = 2$ or $k = 1$?

Proof. $\square$